

Vectors: $\vec{V}$

• Numbers: 1, 2, 3, ... $x, y \leftarrow$ Variables

operators: $\leftarrow$ operation

(i) $1 + 1 = 2$

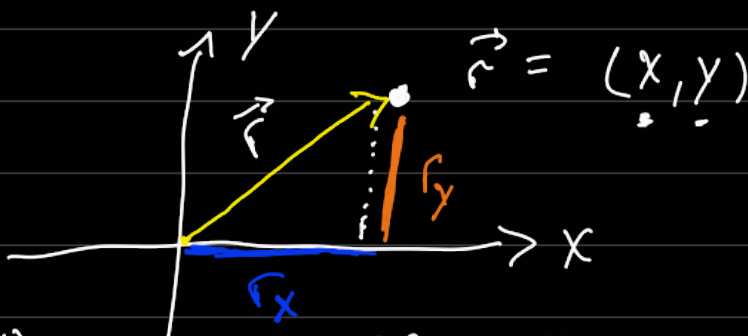
(ii) $|x|$

Vector: two properties

(i) magnitude

(ii) "direction"

ex:



(i) magnitude: "length"

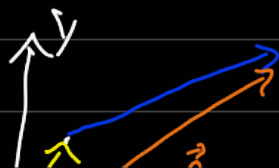
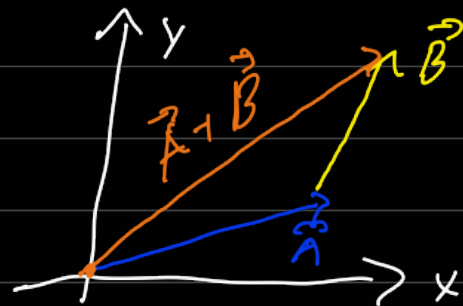
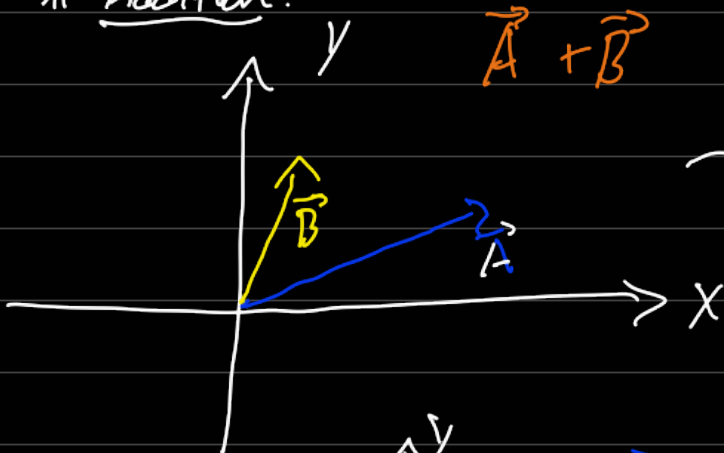


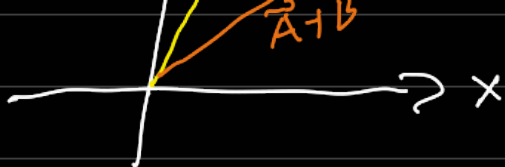
$$a^2 + b^2 = c^2$$

$\rightarrow \boxed{\|\vec{r}\| = c = \sqrt{r_x^2 + r_y^2}}$ (Magnitude / length)

$\left. \begin{array}{l} r_x = a \\ r_y = b \end{array} \right\} \nearrow$

* Addition:





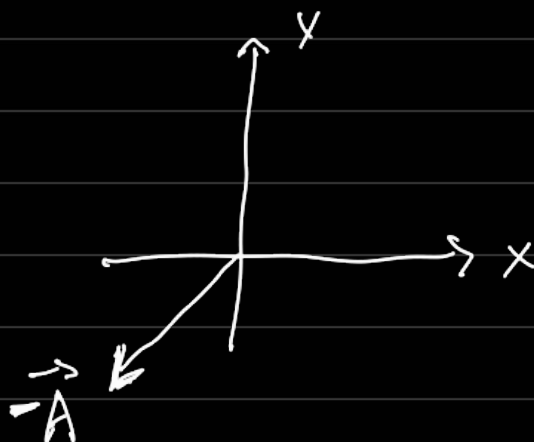
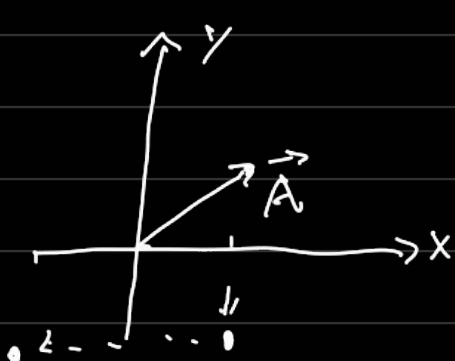
$$\ast \vec{A} = (A_x, A_y) \quad \begin{matrix} x & y \\ \downarrow & \downarrow \end{matrix}$$



$$\vec{A} = (A_x, A_y), \quad \vec{B} = (B_x, B_y)$$

$$\vec{A} + \vec{B} = (A_x, A_y) + (B_x, B_y) = (A_x + B_x, A_y + B_y)$$

Subtraction: $-\vec{A} = (-A_x, -A_y)$



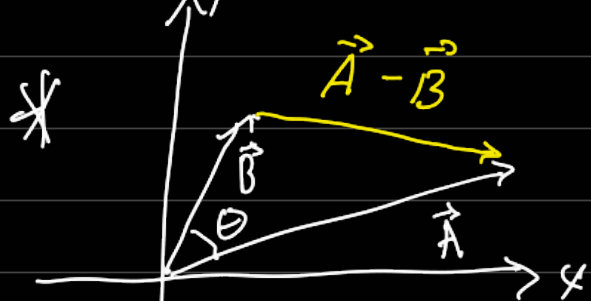
$\ast$ Idea of multiplication:

Introduce the dot-product: Scalar product

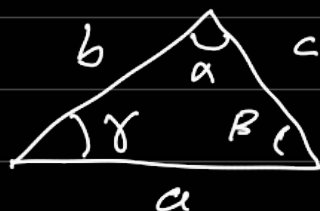
$$\ast \vec{A} \cdot \vec{B} \equiv A_x B_x + A_y B_y$$

Recall: $|\vec{r}|^2 = r_x^2 + r_y^2 = \underbrace{r_x r_x + r_y r_y}_{\vec{r} \cdot \vec{r}}$

$$\ast \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos(\theta)$$



• Law of cosines:



$$c^2 = a^2 + b^2 - 2ab \cos(\gamma)$$

$$c = |\vec{A} - \vec{B}|$$

$$a = |\vec{A}|$$

$$b = |\vec{B}|$$

$$* |\vec{A} - \vec{B}|^2 = |\vec{A}|^2 + |\vec{B}|^2 - 2|\vec{A}||\vec{B}|\cos(\theta)$$

now, scalar product:

$$|\vec{A} - \vec{B}|^2 = (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B})$$

$$= \vec{A} \cdot \vec{A} - \vec{B} \cdot \vec{A} - \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{B}$$

ex: $\vec{A}, \vec{B}, \vec{C}$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

Same as with numbers

$$\bullet x(x+2) = x^2 + 2x$$

$$\bullet (x+2)(x+3) = xy + 3x + 2y + 6$$

$$|\vec{A} - \vec{B}|^2 = |\vec{A}|^2 - 2\vec{A} \cdot \vec{B} + |\vec{B}|^2$$

Since

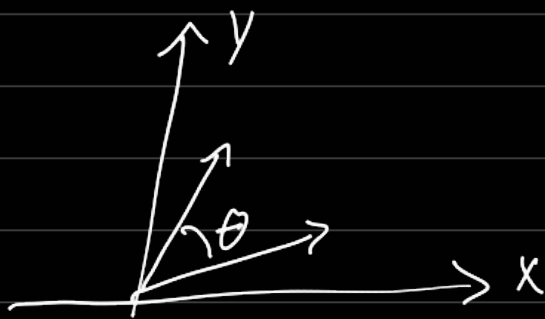
$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\sim 2|\vec{A}||\vec{B}|\cos(\theta) = -2\vec{A} \cdot \vec{B}$$

$$* \boxed{\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|\cos(\theta)}$$

Set these
equal

$$\vec{A} \cdot \vec{A} = |\vec{A}|^2$$



Notation: $\vec{r} = (r_x, r_y)$

often use basis vectors:

Simplest:

$$\hat{x} = \hat{e}_x = (1, 0)$$

$$\hat{y} = \hat{e}_y = (0, 1)$$

$$\mathbb{R}^2 = \text{Span} \{ \hat{x}, \hat{y} \}$$

$$\vec{r} = (r_x, r_y) = (r_x, 0) + (0, r_y)$$

$$= r_x(1, 0) + r_y(0, 1)$$

$$= r_x \hat{e}_x + r_y \hat{e}_y$$

$$\boxed{\vec{r} = r_x \hat{e}_x + r_y \hat{e}_y}$$

$$\begin{aligned} \bullet \mathbb{R}^2 &= \text{Span} \{ \hat{e}_x, \hat{e}_y \} \\ &= \text{Span} \{ 2\hat{e}_x, 3\hat{e}_y \} \end{aligned}$$

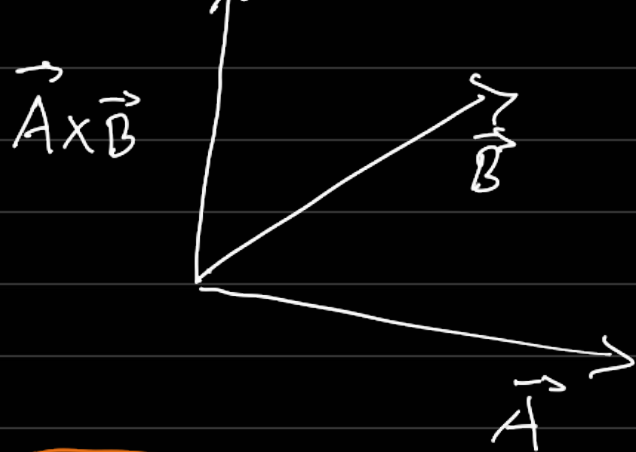
* Cross product: $\vec{A} \times \vec{B}$: vector

• Scalar product gave us a number.

• Cross product gives us a vector

* Move on to 3D: (x, y, z)

$$\vec{A} = (A_x, A_y, A_z), \quad \vec{B} = (B_x, B_y, B_z)$$



A, B on plane

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$$* |\vec{A} \times \vec{B}| = |A| |B| \sin(\theta)$$

$$\vec{A} \times \vec{B} = (A \times B)_x, (A \times B)_y, (A \times B)_z$$

$$\begin{cases} (A \times B)_x = A_y B_z - A_z B_y \\ (A \times B)_y = A_z B_x - A_x B_z \\ (A \times B)_z = A_x B_y - A_y B_x \end{cases}$$

$$(A \times B)_x = A_y B_z - A_z B_y$$

$$(\overbrace{xyz}) \rightarrow (yzx) \rightarrow (zxy)$$

(cyclic property of cross product)

